Debt Ratio Regression: January 2025

Variables used in the regression

1. Debt Ratio = Debt/ (Market Value of Equity + Debt): If you can get market value of debt, use it. Else, use book value of debt.
2. Payout Ratio= Dividends/ Net Income, if Net Income is positve, not available if net income is negative.
3. Expected growth rate in EPS- next 5 years= You can use expected or even historical earnings growth, if you don't have an EPS growth forecast
4. Effective Tax Rate = Effective tax rate in most recent year

US Regression

Model Summary[a]

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.439[b]	.193	.190	2349.30036

a. Broad Group = United States

b. Predictors: (Constant), Company Age, EffTaxRateCorrect, % held by institutions, Expected growth rate in EPS- Next 5 years

ANOVA[a,b,c]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	1650713650	4	412678413	74.771	<.001[d]
	Residual	6921092100	1254	5519212.201		
	Total	8571805750	1258			

a. Broad Group = United States

b. Dependent Variable: DebtotCapCorrect

c. Weighted Least Squares Regression - Weighted by Market Cap (in US $)

d. Predictors: (Constant), Company Age, EffTaxRateCorrect, % held by institutions, Expected growth rate in EPS- Next 5 years

Regression Statistics

Coefficients[a,b,c]

Model		Unstandardized Coefficients B	Unstandardized Coefficients Std. Error	Standardized Coefficients Beta	t	Sig.
1	(Constant)	1.666	1.856		.897	.370
	EffTaxRateCorrect	-.176	.039	-.118	-4.561	<.001
	Expected growth rate in EPS- Next 5 years	-26.978	2.751	-.257	-9.806	<.001
	% held by institutions	18.519	2.292	.209	8.079	<.001
	Company Age	.065	.008	.212	7.905	<.001

a. Broad Group = United States

b. Dependent Variable: DebtotCapCorrect

c. Weighted Least Squares Regression - Weighted by Market Cap (in US $)

Global Regression

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.257[a]	.066	.065	1988.43224

a. Predictors: (Constant), Company Age, % held by institutions, Expected growth rate in EPS- Next 5 years, EffTaxRateCorrect

ANOVA[a,b]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	1369421072	4	342355268	86.588	<.001[c]
	Residual	1.944E+10	4916	3953862.784		
	Total	2.081E+10	4920			

a. Dependent Variable: DebtotCapCorrect

b. Weighted Least Squares Regression - Weighted by Market Cap (in US $)

c. Predictors: (Constant), Company Age, % held by institutions, Expected growth rate in EPS- Next 5 years, EffTaxRateCorrect

Regression Statistics

Coefficients[a,b]

Model		Unstandardized Coefficients		Standardized Coefficients		
		B	Std. Error	Beta	t	Sig.
1	(Constant)	17.419	.803		21.700	<.001
	EffTaxRateCorrect	-.057	.022	-.038	-2.637	.008
	Expected growth rate in EPS- Next 5 years	-20.401	1.480	-.196	-13.786	<.001
	% held by institutions	-2.153	.831	-.037	-2.590	.010
	Company Age	.033	.003	.136	9.583	<.001

a. Dependent Variable: DebtotCapCorrect

b. Weighted Least Squares Regression - Weighted by Market Cap (in US $)

- ***How do I use this regression?***
  Assume that you want to estimate the market debt ratio for a firm with the following characteristics, using the Global regression
  Effective tax rate = 25%
  % held by institutions = 30%
  Company age = 37
  Expected growth rate in EPS = 12%

- ***Predicted Value***
  Expected Debt Ratio = =17.419-0.057 (25) -20.401 (.12) -2.153 (.30) + 0.033 (37) = 14.12 or 14.12%
  If your predicted value is less than zero, your predicted debt ratio is zero.